WORKED SOLUTION

STEP I 2007, Question 5

Pure • Geometry of Polyhedra

STEP I 2007, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q5

The key idea

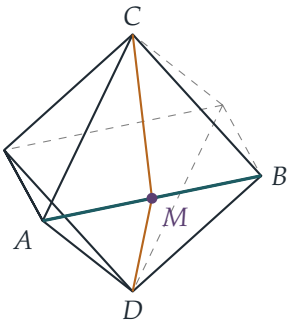
A regular octahedron sits very neatly in coordinates: its six vertices can be the points one unit along each of the positive and negative axes. With that model, both parts are just vectors. The angle between two faces is measured in a plane perpendicular to their common edge, so we use the two lines that run from the midpoint of the edge across each face to the opposite vertex. The face centres are simple averages of three vertices.

Solution

Every regular octahedron is a scaled copy of any other, and neither an angle nor a ratio of volumes changes under scaling. So we may work with the octahedron whose vertices are

$$(\pm 1, 0, 0), \qquad (0, \pm 1, 0), \qquad (0, 0, \pm 1).$$

Any two of these that are not opposite are $\sqrt{2}$ apart, which gives 12 edges, and the faces are the 8 triangles with one vertex on each axis, one for each choice of signs. Every face is equilateral with side $\sqrt{2}$, so this really is a regular octahedron.



Part (i)

Take two faces that share an edge; by symmetry every edge looks the same, so one will do. Let $A = (1,0,0)$, $B = (0,1,0)$, $C = (0,0,1)$ and $D = (0,0,-1)$, so the faces ABC and ABD meet along AB. Let $M = (\frac{1}{2}, \frac{1}{2}, 0)$ be the midpoint of AB.

Ask yourself How do we measure the angle between two planes that meet along a line?

We take a point on the common line and draw, in each plane, the line through it perpendicular to the common line; the angle between those two lines is the angle between the faces. Since each face is equilateral, the median MC of triangle ABC is perpendicular to AB, and so is the median MD of triangle ABD. So the angle we want is $\angle CMD$.

Now

$$\overrightarrow{MC} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\right), \qquad \overrightarrow{MD} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, -1\right),$$

and both are perpendicular to $\overrightarrow{AB} = (-1,1,0)$, as they should be. Each has length squared $\frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$, and

$$\overrightarrow{MC} \cdot \overrightarrow{MD} = \tfrac{1}{4} + \tfrac{1}{4} - 1 = -\tfrac{1}{2}.$$

Hence

$$\cos \angle CMD = \frac{-\frac{1}{2}}{\frac{3}{2}} = -\frac{1}{3},$$

and the angle between the faces is $\arccos\left(-\frac{1}{3}\right)$, about $109.5°$, as required. The cosine is negative, so the angle is obtuse: the octahedron is less sharp along its edges than a cube.

Without coordinates the same calculation is the cosine rule in triangle CMD. With edge length a, the medians are $MC = MD = \frac{\sqrt{3}}{2}a$ and C, D are opposite vertices: the four points A, C, $(-1,0,0)$, D form a square of side a, so $CD = \sqrt{2}\,a$. Then

$$2a^2 = \tfrac{3}{4}a^2 + \tfrac{3}{4}a^2 - 2 \cdot \tfrac{3}{4}a^2 \cos \angle CMD \quad \Longrightarrow \quad \cos \angle CMD = -\tfrac{1}{3}.$$

Part (ii)

The centre of a face is the average of its three vertices. A face has one vertex $(\pm 1, 0, 0)$, one $(0, \pm 1, 0)$ and one $(0, 0, \pm 1)$, so its centre is

$$\left(\pm\tfrac{1}{3}, \pm\tfrac{1}{3}, \pm\tfrac{1}{3}\right),$$

and the eight choices of signs give the eight centres. These are exactly the vertices of a cube with side $\frac{2}{3}$, so

$$V_{\text{cube}} = \left(\tfrac{2}{3}\right)^3 = \tfrac{8}{27}.$$

The octahedron is two square-based pyramids glued together along the square with vertices $(\pm1, 0, 0)$, $(0, \pm1, 0)$. That square has diagonals of length 2, so its area is $\frac{1}{2} \cdot 2 \cdot 2 = 2$, and each pyramid has height 1. Therefore

$$V_{\text{oct}} = 2 \times \tfrac{1}{3} \times 2 \times 1 = \tfrac{4}{3}.$$

The ratio is

$$\frac{V_{\text{oct}}}{V_{\text{cube}}} = \frac{4/3}{8/27} = \frac{9}{2},$$

so the octahedron's volume is to the cube's as $9 : 2$.

Checks

- $\arccos(-\frac{1}{3}) \approx 109.47°$. Repeating the midpoint calculation numerically for all 12 edges of the coordinate model gives $\cos = -\frac{1}{3}$ every time.
- The octahedron is $|x| + |y| + |z| \leq 1$. A Monte Carlo estimate of its volume from 10^6 random points in $[-1, 1]^3$ gave 1.330, against the exact $\frac{4}{3}$.
- The cube is inside the octahedron (each centre satisfies $|x| + |y| + |z| = 1$), so the ratio must be greater than 1. And $\frac{9}{2}$ is.

Common slips

- Taking the angle between the outward normals, $(1, 1, 1)$ and $(1, 1, -1)$, which gives $\arccos \frac{1}{3}$. The angle between the faces themselves is the supplement of that.
- Measuring the angle along lines that are not perpendicular to the common edge, such as two edges CA and DA. That angle is $90°$ here, and it isn't the angle between the faces.
- Thinking the cube's side is the distance between the centres of two opposite faces. That distance, $\frac{2}{\sqrt{3}}$, is a space diagonal of the cube.

Beyond STEP

The cube and the octahedron are *dual* polyhedra: joining the face centres of either gives the other, with faces and vertices swapping roles. Duality is the first sign that the symmetry groups of the two solids are the same group, of order 48, which you'll meet in a first course on groups. For the symmetry groups of the Platonic solids, try Armstrong, *Groups and Symmetry*.

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